

ECSE-4040

Digital Electronics

Lecture 8

Prof. T. Paul Chow

Bldg. CII, Room 6111

Rensselaer Polytechnic Institute

Troy, NY 12180-3590

Tel. (518)276-2910

FAX (518)276-2990

e-mail: chowt@rpi.edu

<https://lms.rpi.edu>



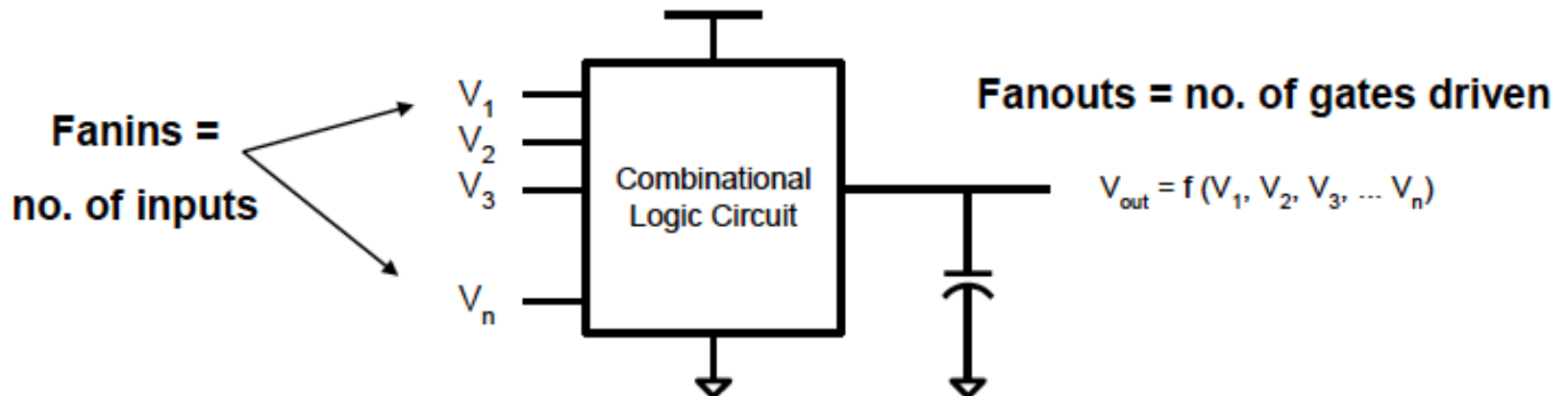
Lecture Outline

- Combinational MOS Logic Circuits
- Static Logic Gates
 - “Beta” Ratio
 - NAND and NOR Sizing
 - VTC and Noise Margins
- Complex Logic Circuits
- Pseudo-NMOS Logic



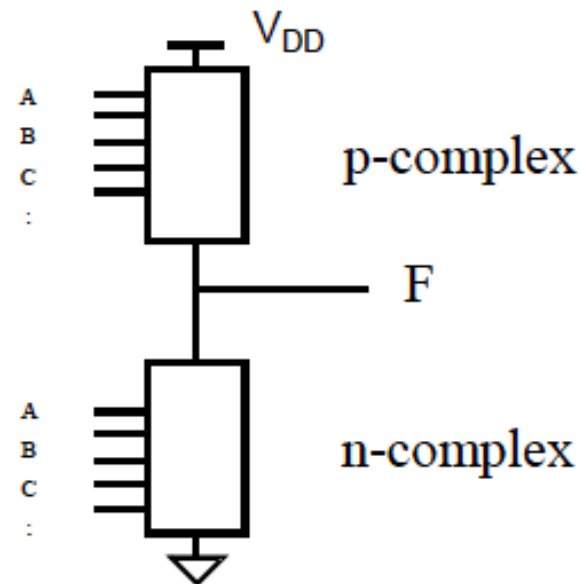
Combinational MOS Logic

- Now that we understand the logic abstraction and the properties of valid logic gates, we can consider the issues of design basic building blocks of digital systems
- Typical combinational gate is a multiple input – single output system
- Performs Boolean operations on multiple input variables, drives one or more gates
- Design parameters and considerations:
 - Propagation delay
 - Static and dynamic power
 - Area
 - Noise margins (VTC)



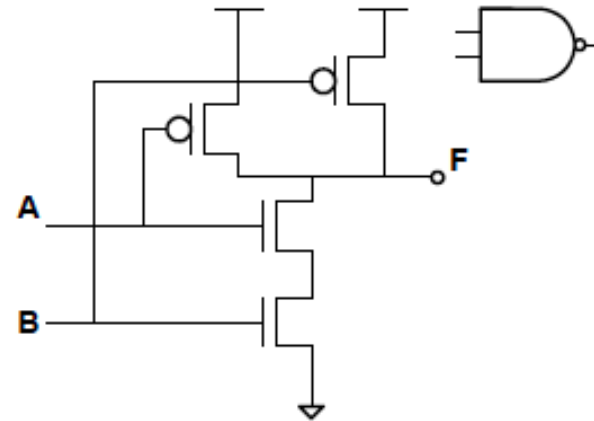
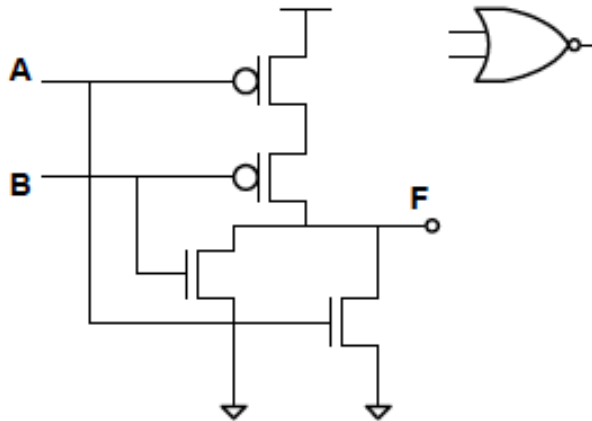
Pull-up and Pull-down Network

**Basic Structure
Of all CMOS logic gates**



- PMOS pull-up and NMOS pull-down networks are duals of each other
- Configuration of pull-up and pull-down networks create a current connection from the output to *either* V_{DD} or GND , based on the inputs
- PMOS devices have lower drive capability and thus require wider devices to achieve the same on-resistance as its pull-down counterpart

Static CMOS Logic Gates



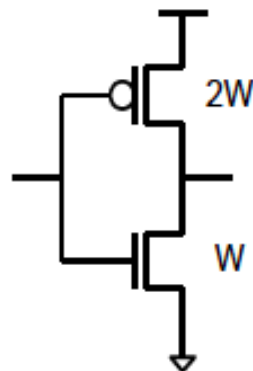
- These are the most common type of static gates
- Can implement any Boolean expression with these two gates
- Why is static CMOS so popular?
 - It's very robust!
 - it will eventually produce the right answer
 - Power, shrinking V_{DD} , more circuit noise, process variations, etc. limit use of other design styles

More CMOS Logic Properties

- Full complementary
- Low static power dissipation
- Output swings full rail – from V_{DD} (V_{OH}) to Gnd (V_{OL})
- Works fine at low V_{DD} voltages
 - But lower V_{DD} → less current → slower speed
- Combinational operation
 - Feed it some inputs, wait some delay, result comes out
 - No clocks required for normal operation
- Moderately good performance
 - Drive strength is proportional to transistor size
 - Large loads require large W
- Dual logic networks for N and P-Channel Devices

“Beta Ratio” for Static Gates

- The ratio between the NMOS and PMOS device is called the beta ratio
- We need to size CMOS static gates to deliver a target speed. But how?
- Start by sizing the inverter to deliver the target speed, then map size to gate
- Suppose:
 $R_{eqp} \text{ (PMOS)} \approx 2.5 R_{eqn} \text{ (NMOS)}$ under identical conditions
 (actually $R_{eqp}=30k\Omega$ and $R_{eqn}=12.5k\Omega$)
- Then ratio between PMOS:NMOS should be 2.5:1
- Beta ratio sets:
 - Switching point of the gate output drive
 - Input capacitance
 - L-to-H vs. H-to-L transition times
 - Usually find a 2:1 ratio in CMOS inverter



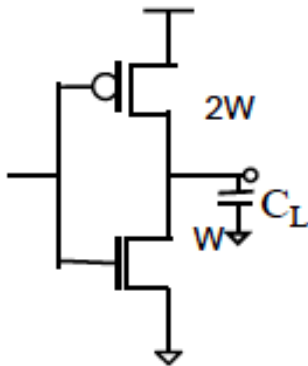
Is 2:1 the right ratio?

For equal rise and fall times. 2.5:1

How about for minimum delay? 1.7:1

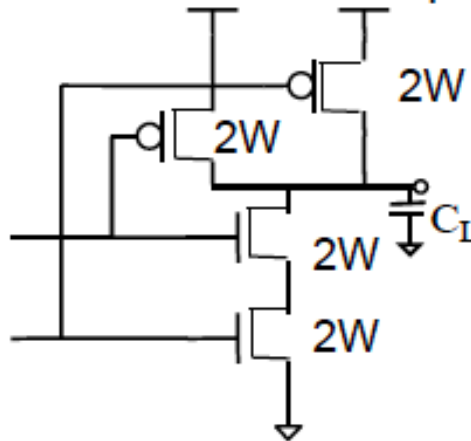
NAND and NOR Sizing

First Design Inverter

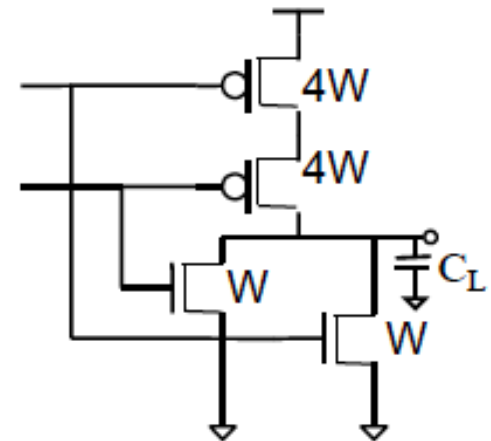


(a) Inverter

Then Map Results to Gates



(b) NAND

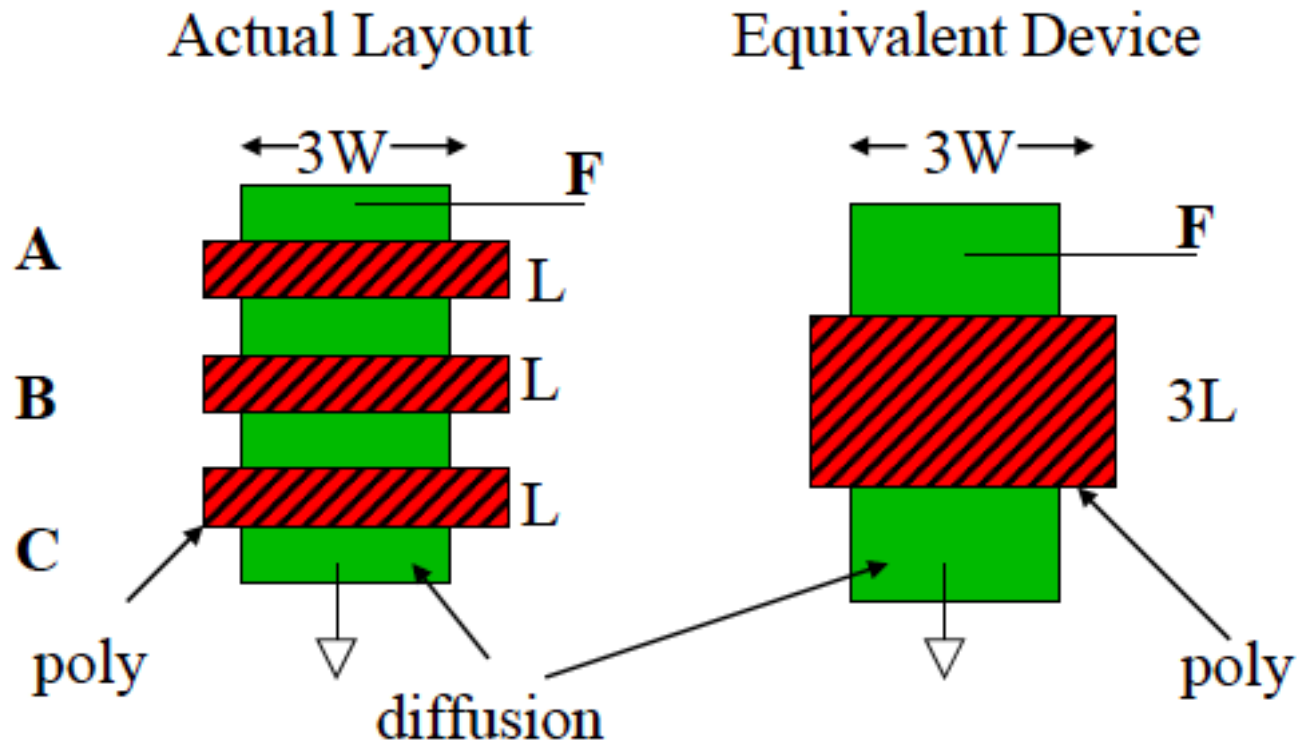


(c) NOR

- Drive strength determined by device widths - W (assume L is minimum size)
- For the moment, consider only C_L (we are ignoring the device self-capacitance)
- Pick the right sizes for the basic inverter and then assign values to gates
- What does that mean for parallel and series combinations?
 - For parallel transistors, direct mapping from inverter
 - For series transistors, need to compute equivalent sizes

Equivalent Sizes

- Consider a three input NAND gate (NMOS portion only):



Equivalent Transistor Width

- R_{ON} of transistor is proportional to channel width, W
- Equivalent channel width:

$$W_{eq} = \frac{W_1 W_2 W_3}{W_1 W_2 + W_2 W_3 + W_3 W_1} \quad (\text{series stack, all devices on})$$

- If $W_1=W_2=W_3=3W_n$, then $W_{eq} = W_n$

$$W_{eq} = W_1 + W_2 + W_3 \quad (\text{parallel devices, all devices on})$$

- Worst case, only one pull-down transistor is on for NOR gate

$$\text{If } W_1=W_2=W_3=3W_n, \text{ then } W_{eq} = W_n$$

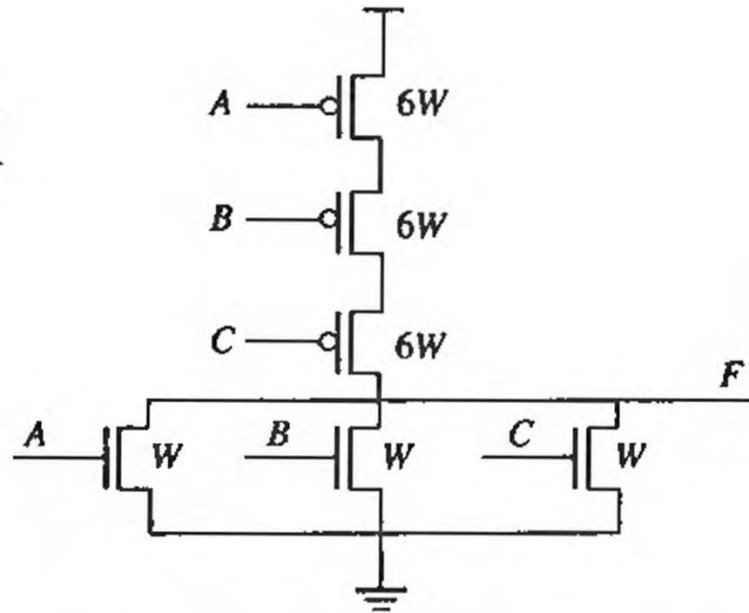
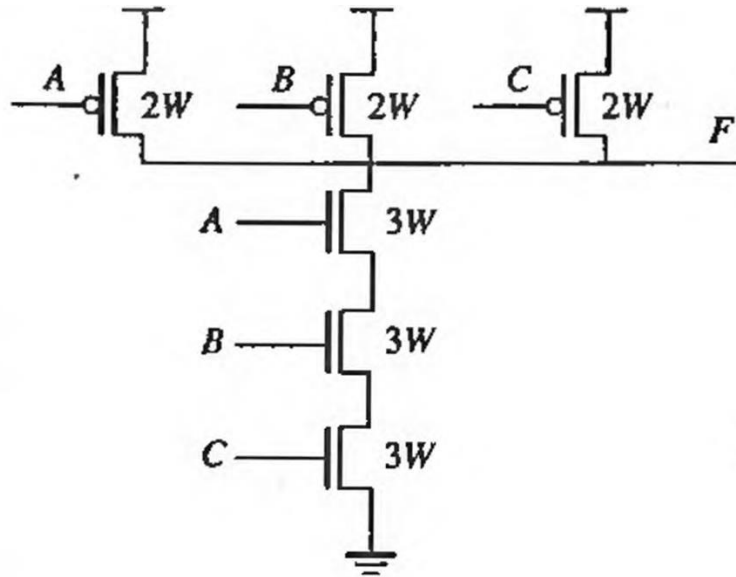
HJS Ex. 5.1

3-Input NAND and NOR Gate Sizing

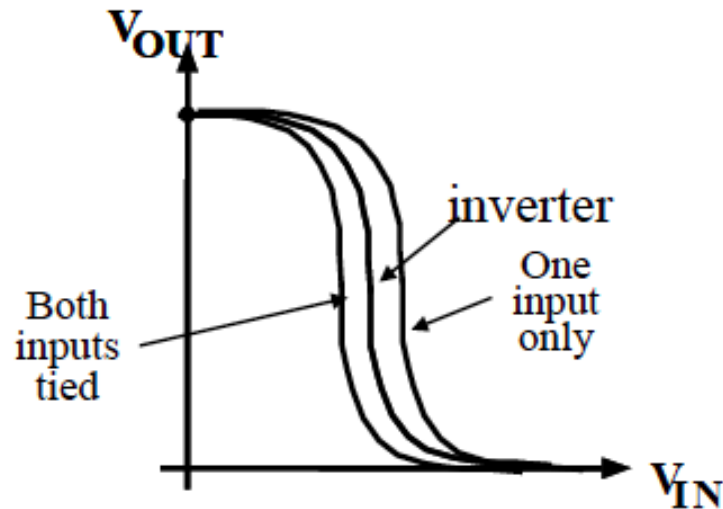
Problem:

Determine the device sizes for 3-input NAND and NOR gates in conventional CMOS. Assume that the basic inverter is sized according to Figure 5.2a and that the goal of the design is to have NAND3 and NOR3 gates with the same delay characteristics as the inverter.

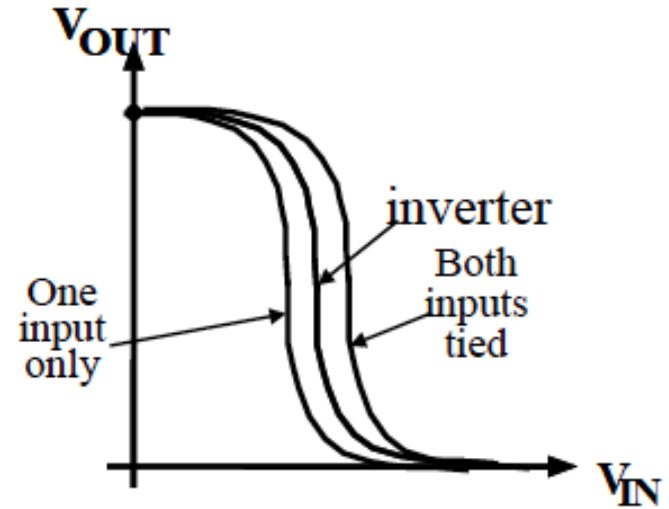
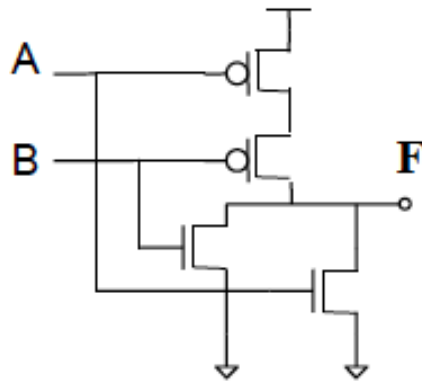
HJS Ex. 5.1



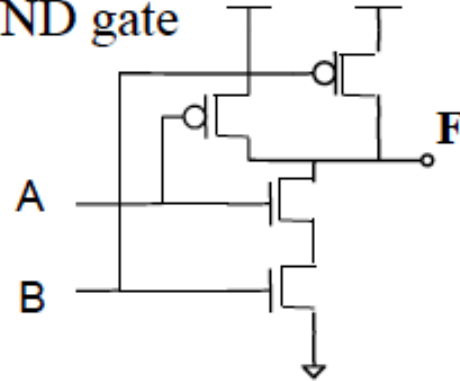
VTC and Noise Margins



NOR gate



NAND gate



HJS Ex. 5.2

SPICE Example of VTC for NAND Gate in 0.18 μm Technology

Problem:

- (a) Compute the switching threshold, V_S , of a 2-input NAND gate shown in Figure 5.10 under two conditions: with one input tied to V_{DD} while the other input is swept from 0 to V_{DD} and a second case where both inputs are tied together. Assume 0.18 μm technology parameters with all transistor widths set to 400 nm.
- (b) Using SPICE, compare the VTC curves of a 2-input NAND for the two cases above and compare the switching threshold results. Also include the two inputs, A and B, switching separately. Why are the results slightly different for each input?



HJS Ex. 5.2

- (a) When one input is high and the other is varied from 0 to V_{DD} , it is equivalent to having a pull-up with $W_p = 400$ nm and a pull-down of $W_N = 400$ nm (treat the other one as a very small resistor):

$$W_N = 400 \text{ nm}, \quad W_p = 400 \text{ nm:}$$

$$X = \sqrt{\frac{W_N/E_{CN}L_N}{W_p/E_{CP}L_p}} = \sqrt{\frac{W_N E_{CP}}{W_p E_{CN}}} = \sqrt{\frac{(400)(24)}{(400)(6)}} = 2.0$$

$$V_S = \frac{1.8 - 0.5 + (0.5)2.0}{1 + 2.0} = 0.77 \text{ V}$$

When both inputs are varied from 0 to V_{DD} , it is equivalent to having a pull-up with $W_p = 800$ nm and a pull-down of 200 nm:

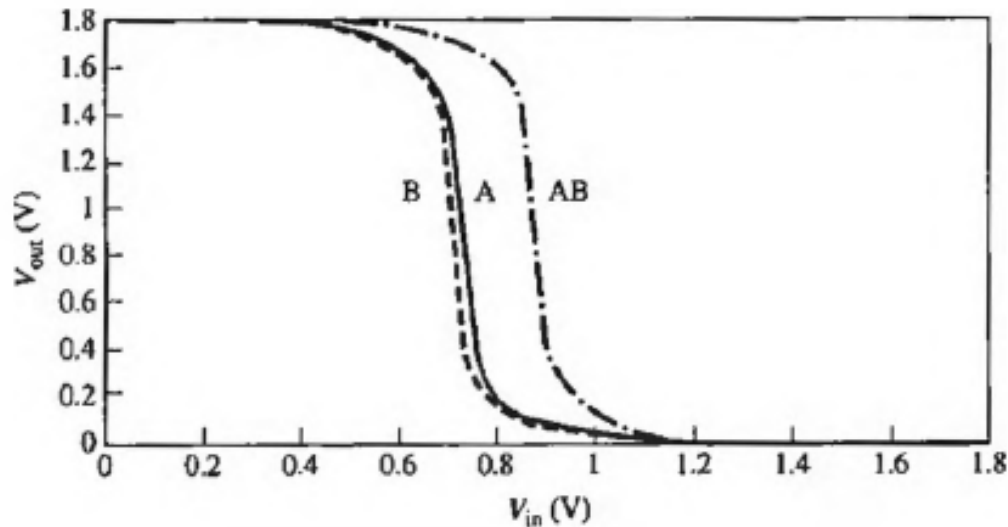
$$W_p = 800 \text{ nm}, \quad W_N = 200 \text{ nm:}$$

$$X = \sqrt{\frac{W_N/E_{CN}L_N}{W_p/E_{CP}L_p}} = \sqrt{\frac{W_N E_{CP}}{W_p E_{CN}}} = \sqrt{\frac{(200)(24)}{(800)(6)}} = 1.0$$

$$V_S = \frac{1.8 - 0.5 + (0.5)1.0}{1 + 1.0} = 0.9 \text{ V}$$

HJS Ex. 5.2

The following results are produced by SPICE:



As expected, the curve with both inputs tied is to the right of the curve for single inputs. From SPICE, $V_S = 0.745$ V for input A switching, $V_S = 0.721$ V for input B switching, and $V_S = 0.883$ V for the case where the two inputs are tied together. These are close to the hand calculated values. The hand calculations tend to produce slightly higher values due to the approximations made to simplify the expression for V_S . The reason why input A produces a slightly higher value of V_S is that the corresponding transistor experiences body effect during the switching process.

Complex Logic Circuits

- Ability to easily build complex logic gates is one of the most attractive features of MOS logic circuits
- Design principle of pull-down network
 - OR operations are performed by parallel connected drivers
 - AND operations are performed by series connected drivers
 - Inversion is provided by the nature of MOS circuit operation
- Don't get too carried away...Use this knowledge wisely
 - Remember that complex functions don't have to be implemented with a single gate
 - Can break up very complicated Boolean expressions into a cascade of gate stages
- We will use DeMorgan's Law to build dual networks

Review of DeMorgan's Law

$$\begin{array}{c} a \\ b \end{array} \text{ AND } = \begin{array}{c} a \\ b \end{array} \text{ OR } \quad \overline{ab} = \overline{a} + \overline{b}$$

$$\begin{array}{c} a \\ b \end{array} \text{ OR } = \begin{array}{c} a \\ b \end{array} \text{ AND } \quad \overline{\overline{a} \overline{b}} = \overline{\overline{a} + \overline{b}}$$

- De Morgan's theorem:

The complement of any logic function is found by complementing all input variables and replacing all AND operations with OR and all OR operations with AND

- Use De Morgan's law to find the complement of a function for the pull-down network (if needed)
- Use Duality to find the pull-up network

Complex CMOS Gate Design Example

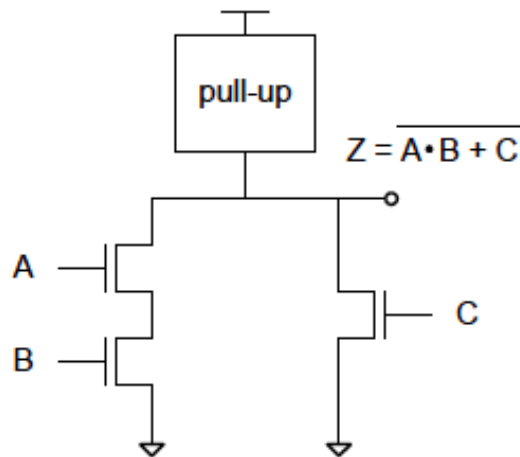
- Implement an AND-OR-INVERT (AOI) function

$$Z = \overline{A \cdot B + C}$$

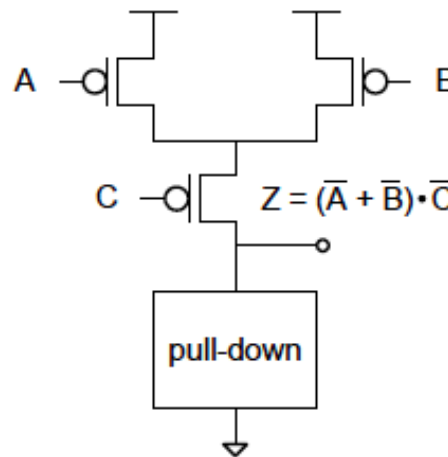
- Get the expression into forms that enable easy implementation of pull-up and pull-down networks

$$\overline{Z} = (A \cdot B + C)$$

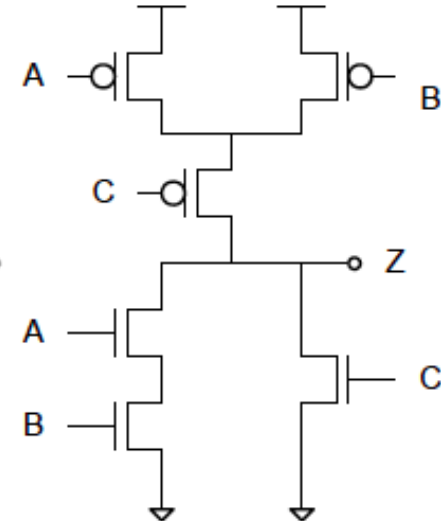
$$Z = (\overline{A} + \overline{B}) \cdot \overline{C}$$



Complement of Z



Dual of Complement of Z



Combine Results

HJS Ex. 5.3

Complex CMOS Gate Circuits

Problem:

Implement $F = \overline{AB} + \overline{CD}$ as a single-stage complex CMOS gate and a pseudo-NMOS gate.



HJS Ex. 5.3

Solution:

For the CMOS gate: first complement F to obtain the n -complex function: $\bar{F} = AB + CD$. This implies that we need two series NMOS paths in parallel with one another. One series path implements AB while the other implements CD .

Next, take the dual of the complement of F :

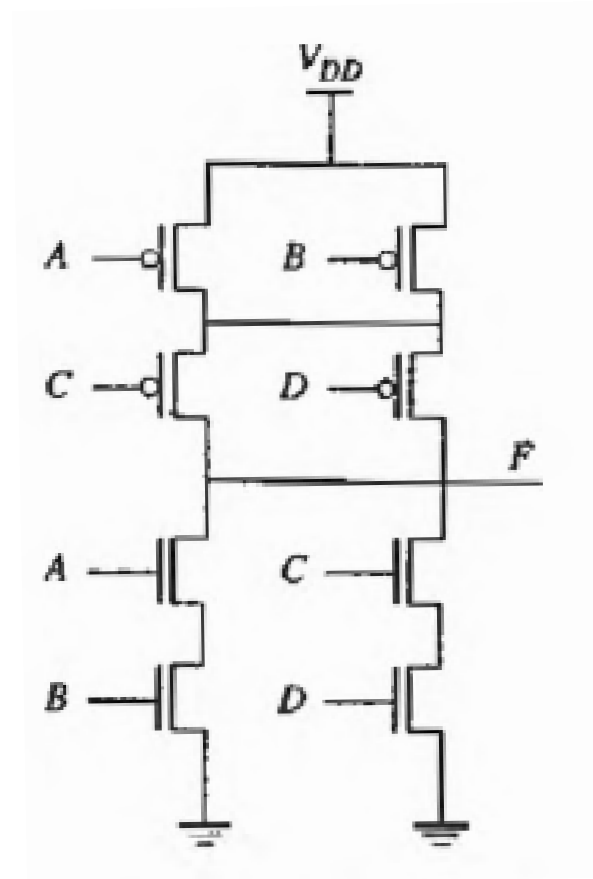
$$\bar{F}|_{dual} = (A + B)(C + D)$$

This is the function implemented in the p -complex. The same expression would be obtained if we applied DeMorgan's Law to F and then complemented each literal. This implies that we need a series connection of two parallel devices, one with A or B and the other with C or D .

HJS Ex. 5.3

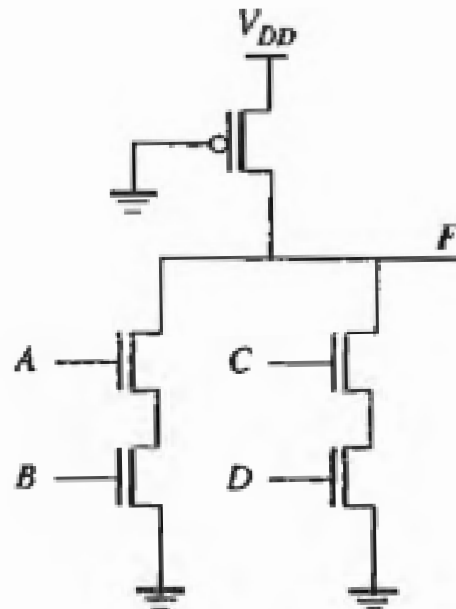
CMOS Implementation

$$F = \overline{AB + CD}$$

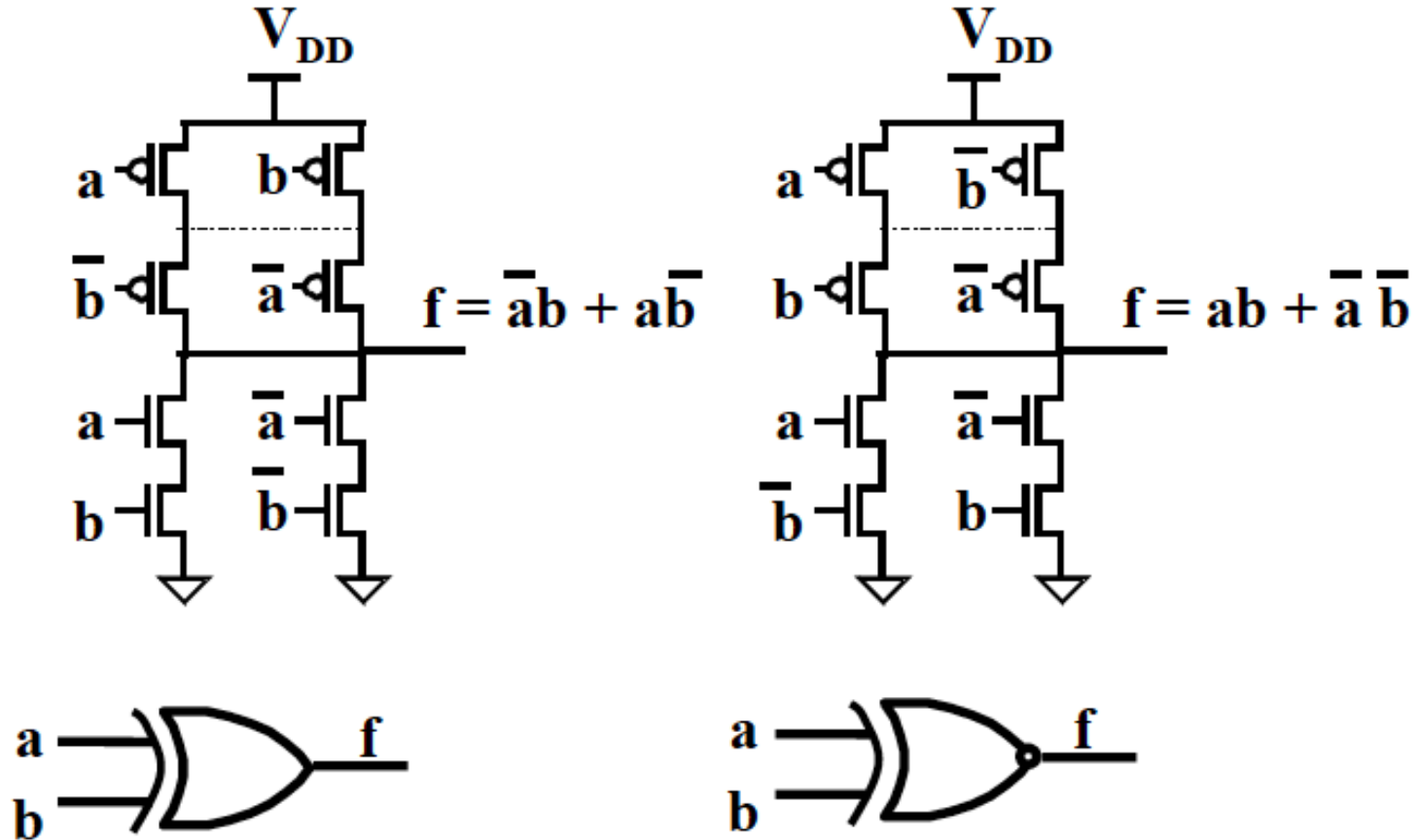


HJS Ex. 5.3

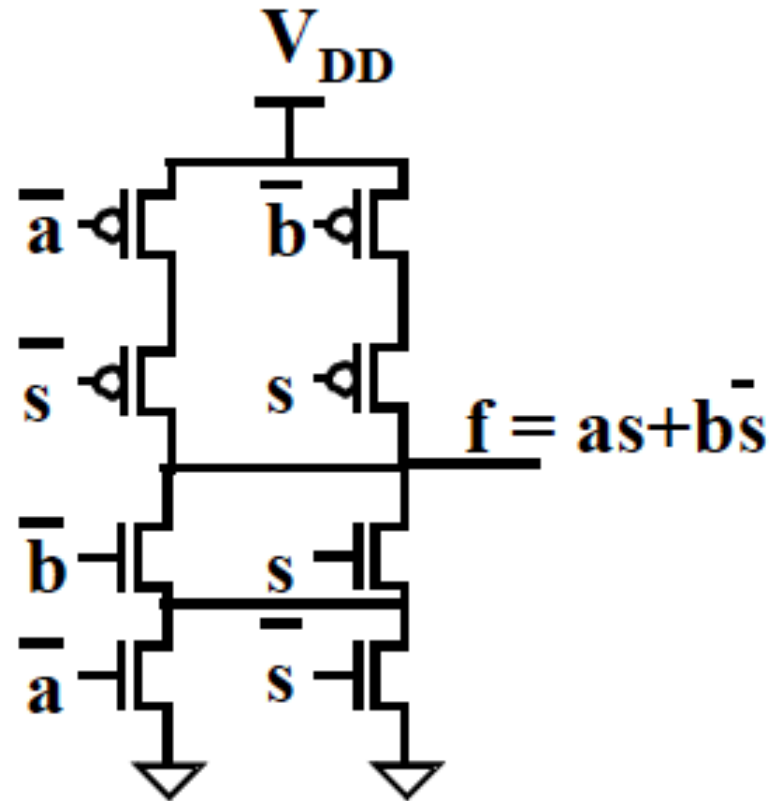
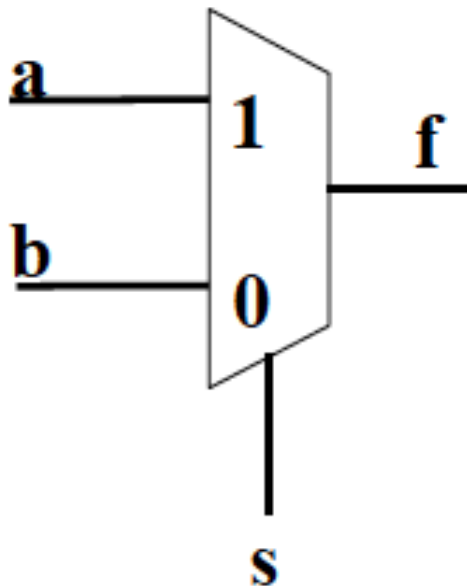
For the pseudo-NMOS gate, only the first part is needed to construct the n -complex. The pull-up network is, of course, only a single PMOS device. Clearly, this is more efficient than the CMOS case in terms of area.



XOR and XNOR Gates



CMOS Multiplexer

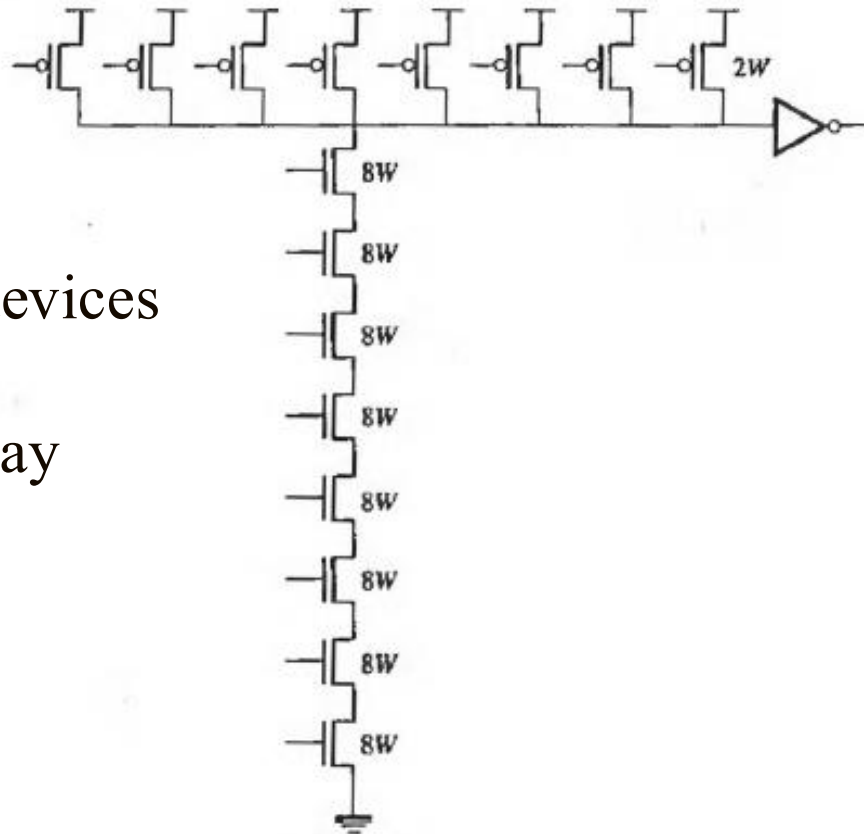


The Bad News

- Slows down dramatically for large fan-ins due to long series stack of transistors
 - Fan-ins = number of inputs
 - PMOS stacks are worse than NMOS series stacks
- Large number of transistors
 - $2n$ devices for n -input NAND
 - At least 2 devices per input
- Bigger layout
 - N^+ and P^+ spacing rule and well spacing rule
 - Large device sizes required to counteract series stack
- Limit fan-in to 3 or 4...or delay and area will be too large

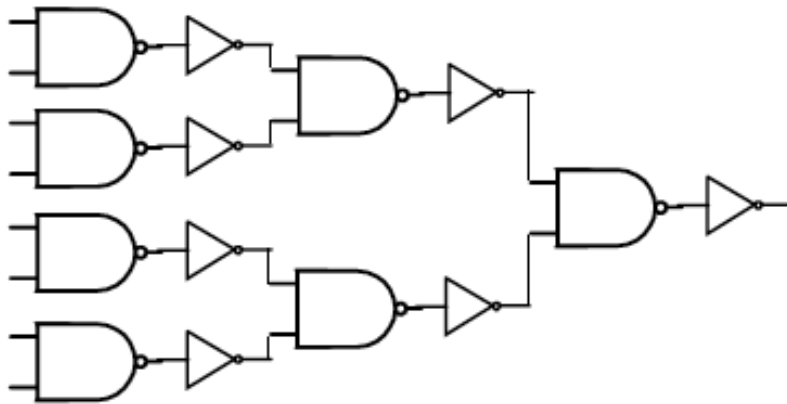


Eight-Input AND Gate

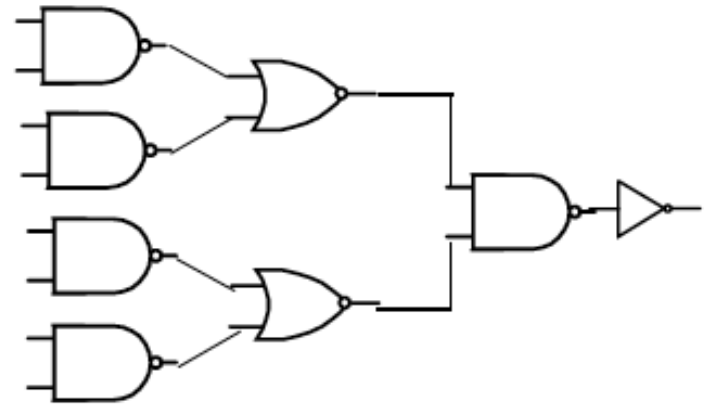


- 8 series NMOS devices too large
- Unacceptable delay due to RC ladder circuit

Multi-Level Logic Implementations



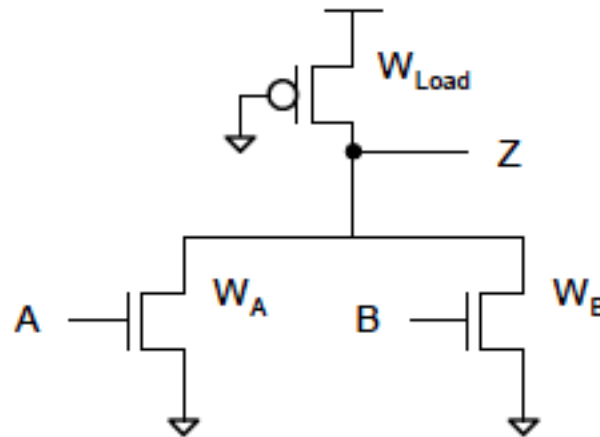
NAND2-INV-NAND2-INV



NAND2-NOR2-NAND2-INV

- There are many more options to try
- Which is the best? We need a quick way of answering this question

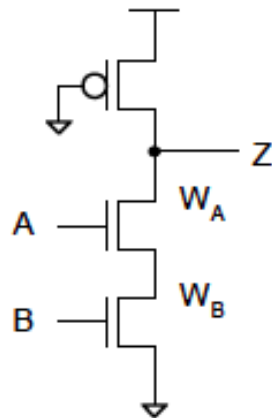
Pseudo-NMOS Logic – NOR Gate



A	B	Z
0	0	1
0	1	0
1	0	0
1	1	0

- Design issues:
 - Sizing Ratio
 - Ratio pull-up to pull-down (V_{OL} & V_{OH})
 - Propagation delay
 - Subthreshold current can degrade V_{OH} slightly
 - V_{OL} decreases as more devices turning on

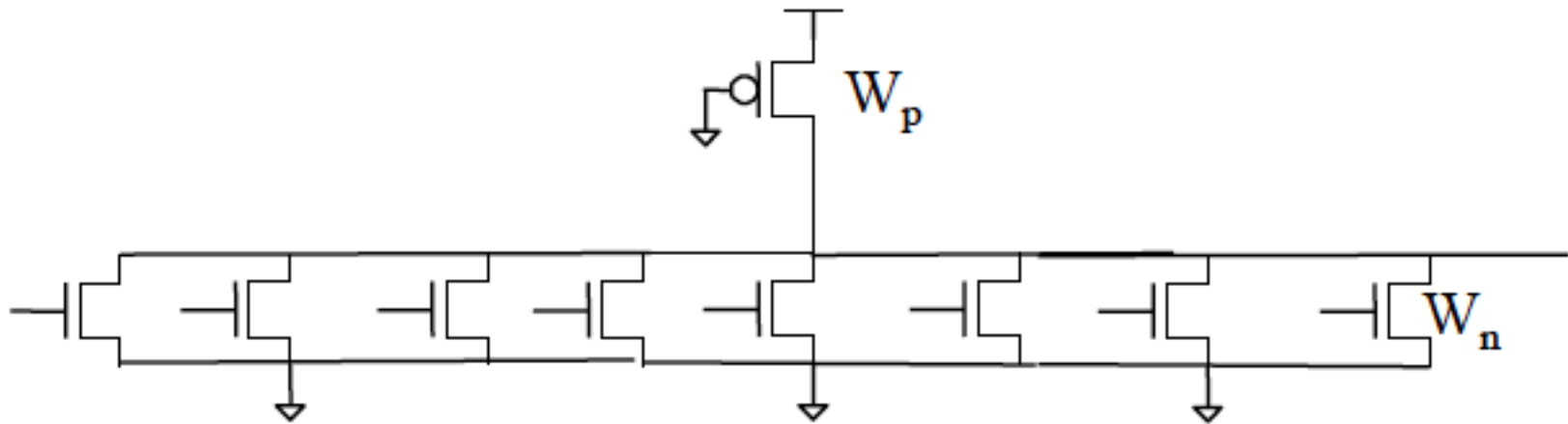
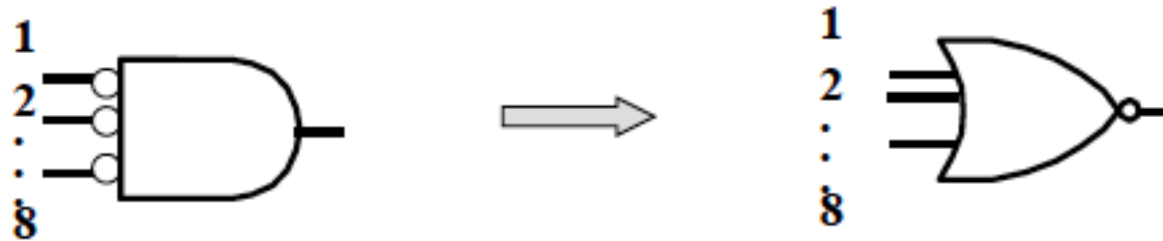
Pseudo-NMOS Logic – NAND Gate



A	B	Z
0	0	1
0	1	1
1	0	1
1	1	0

- Issues
 - Sizing Ratio
 - Need to make pull-down devices wider
 - Parasitic cap goes up with bigger devices
 - Lower devices in stack slower compared to upper ones because they see more capacitance

AND8 Option – Use Pseudo-NMOS



Properties of Static Pseudo-NMOS Gates

- DC power
 - Always conducting current when output is low
- V_{OL} and V_{OH} depend on sizing ratio and input states
- Poor low-to-high transition
- Large fan-in NAND gates tend to get big due to ratioing
- With transistor count increasing, power consumption is too high
 - Cannot use this approach for all gates on the chip
- But what are its advantages?
 - Good for wide NOR structures (e.g., memory decoder)
 - Smaller number of transistors (area) per logic function